

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 29 — An Introduction to A2 Organic Chemistry

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Chirality & Enantiomerism · Optical Activity & Polarimetry · Racemic Mixtures · Stereochemical Reaction Mechanisms
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

- | | |
|--|---|
| 29.1 Chirality, Asymmetric Carbons & Enantiomers | — Definition of chiral center (asymmetric carbon atom with four different groups); enantiomers as mirror images |
| 29.2 Optical Activity, Polarimetry & Racemic Mixtures | — Plane-polarised light; operation of a polarimeter; rotation of plane-polarised light in opposite directions |
| 29.3 Stereochemical Outcomes of Organic Reactions | — Planar trigonal intermediates; nucleophilic addition of HCN to planar aldehydes/ketones yielding racemic mixtures |
| 29.4 Pharmaceutical Relevance of Chirality | — Biological receptor stereospecificity; differing pharmacological activities of enantiomers (e.g. thalidomide) |
| High-Frequency Core Repeats (Q41–Q50) | — The 10 most frequently tested Cambridge Paper 4 questions on A2 Organic Foundations from the last decade |

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Optical Isomerism of 2-Hydroxypropanoic Acid (Lactic Acid) — 9701/42/M/J/23/Q7 [6 Marks]

HARD · Syllabus 29.1

2-Hydroxypropanoic acid (lactic acid), $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, is produced in muscle tissue during anaerobic respiration. It exists as a pair of optical enantiomers, as illustrated in Fig. 1.1.

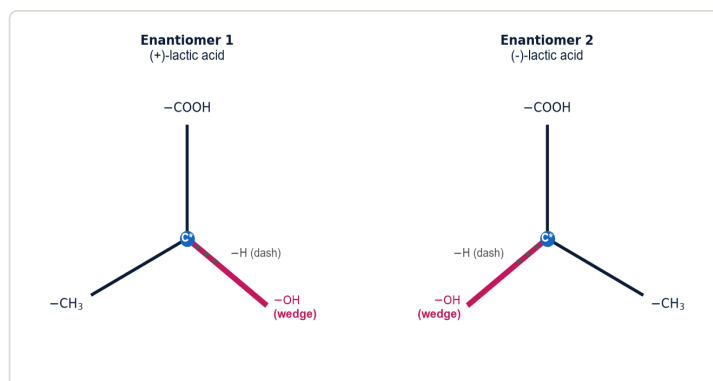


Fig. 1.1: 3D non-superimposable tetrahedral enantiomers of lactic acid reflecting across a mirror plane.

- (a) Identify the chiral center in lactic acid by writing its structural formula and marking the asymmetric carbon with an asterisk (*). [2]

.....

.....

- (b) Draw 3D wedge-and-dash diagrams of the two enantiomers of lactic acid, showing the reflection across a vertical mirror plane. [2]

.....

.....

.....

- (c) State how the two enantiomers can be distinguished experimentally, and describe what is observed when equal amounts of both enantiomers are mixed. [2]

.....

.....

2 Nucleophilic Addition of HCN to Ethanal & Racemic Product Formation — 9701/41/M/J/23/Q7 [6 Marks]

HARD · Syllabus 29.3

Ethanal reacts with hydrogen cyanide in the presence of a catalytic amount of NaCN to form 2-hydroxypropanenitrile:



The product exhibits optical activity when separated, but the reaction mixture is optically inactive.

- (a) Outline the two-step nucleophilic addition mechanism for this reaction, using curly arrows and showing full charges. [3]

.....

.....

.....

- (b) Explain why the product 2-hydroxypropanenitrile formed in this reaction is a racemic mixture. [2]

.....

.....

- (c) Explain why propanone reacting with HCN produces a compound that has no optical isomers. [1]

.....

.....

3 Stereochemical Comparison of S_N1 and S_N2 Reaction Pathways — 9701/42/O/N/23/Q7 [6 Marks]

HARD · Syllabus 29.3

The hydrolysis of halogenoalkanes can proceed via an S_N1 or S_N2 mechanism. Optically active (R)-2-bromobutane and optically active (R)-2-bromo-2-methylbutane are hydrolysed with aqueous sodium hydroxide.

- (a) Describe the stereochemical outcome of the S_N2 hydrolysis of (R)-2-bromobutane, explaining the mechanism in terms of backside attack. [3]

.....

.....

.....

- (b) Describe the stereochemical outcome of the S_N1 hydrolysis of a tertiary chiral bromoalkane, explaining why racemisation occurs. [3]

.....

.....

.....

4 Pharmaceutical Significance of Chirality: Thalidomide & Ibuprofen — 9701/41/O/N/23/Q7 [6 Marks]

HARD · Syllabus 29.4

Many pharmaceutical drugs contain one or more chiral carbon atoms and are administered as single enantiomers or racemic mixtures.

- (a) Explain why two enantiomers of a drug molecule can have vastly different biological effects in the human body. [2]

.....

.....

- (b) Thalidomide was prescribed in the 1950s as a racemate. State the therapeutic effect of one enantiomer and the tragic effect of the other. [2]

.....

.....

- (c) State two economic or health advantages of manufacturing and marketing a drug as a pure single enantiomer rather than a racemic mixture. [2]

.....

.....

5 Multiple Chiral Centers: Tartaric Acid & Meso Compounds — 9701/42/M/J/22/Q7 [6 Marks]

HARD · Syllabus 29.1

Tartaric acid, HOOC-CH(OH)-CH(OH)-COOH, contains two asymmetric carbon atoms.

- (a) Calculate the maximum theoretical number of stereoisomers for a compound with two chiral centers. [1]
-
-
- (b) Tartaric acid exists as only three stereoisomers: a pair of enantiomers and a meso compound. Draw the meso compound and explain why it is optically inactive despite containing two chiral centers. [3]
-
-
-
-
- (c) Explain what is meant by diastereomers, and compare their physical properties with those of enantiomers. [2]
-
-
-

6 Polarimetry Instrumentation & Specific Rotation — 9701/41/M/J/22/Q7 [6 Marks]

HARD · Syllabus 29.2

The optical rotation of a sugar solution is measured using a polarimeter.

- (a) Explain what is meant by plane-polarised light, and describe how unpolarised light is converted into plane-polarised light. [2]
-
-
-
- (b) Describe the main components and operation of a polarimeter. [2]
-
-
-
- (c) A 10.0% solution of an optically active compound in a 10.0 cm tube gives an observed rotation of +6.5°. Calculate its specific optical rotation [α], and predict the rotation if the path length is doubled. [2]
-
-
-

7 Stereochemistry of Alkene Addition Reactions: Bromination of Ethene vs But-2-ene — 9701/42/O/N/22/Q7 [6 Marks]

HARD · Syllabus 29.3

Electrophilic addition of bromine to alkenes proceeds via a cyclic bromonium ion intermediate.

- (a) Write the mechanism for the reaction of propene with bromine, showing curly arrows and the cyclic bromonium ion. [3]
-
-
-
-
- (b) Explain why 1,2-dibromopropane formed in this reaction possesses an asymmetric carbon atom, and identify it. [2]
-
-
-

- (c) State whether the 1,2-dibromopropane formed from propene is optically active or optically inactive, giving your reasoning. [1]
-
-

8 D- and L-Amino Acids & Peptide Bond Stereochemistry — 9701/41/O/N/22/Q7 [6 Marks]

HARD · Syllabus 29.1

All 20 standard naturally occurring amino acids (except glycine) are chiral and exist naturally almost exclusively as L-enantiomers.

- (a) Explain why glycine, $\text{H}_2\text{NCH}_2\text{COOH}$, is the only amino acid that does not exhibit optical activity. [2]
-
-

- (b) Draw the 3D structure of L-alanine, $\text{CH}_3\text{CH}(\text{NH}_2)\text{COOH}$, showing the tetrahedral geometry around the chiral carbon. [2]
-
-
-

- (c) Explain why enzymes composed of L-amino acids can recognise and metabolise only D-glucose and not L-glucose. [2]
-
-
-

9 Asymmetric Synthesis & Chiral Catalysts — 9701/42/M/J/21/Q7 [6 Marks]

HARD · Syllabus 29.4

Modern pharmaceutical synthesis increasingly utilizes asymmetric synthesis to produce single enantiomers directly.

- (a) Define the term asymmetric synthesis (enantioselective synthesis). [2]
-
-

- (b) Describe two methods used to obtain a pure enantiomer from a reaction that initially produces a racemic mixture. [2]
-
-
-

- (c) Explain how the use of a chiral transition metal catalyst (such as Noyori ruthenium catalysts) produces a single enantiomer. [2]
-
-
-

10 Identifying Chiral Centers in Complex Natural Products: Menthol — 9701/41/M/J/21/Q7 [6 Marks]

HARD · Syllabus 29.1

Menthol is a monocyclic terpene alcohol found in peppermint oil.

Its structure is 2-isopropyl-5-methylcyclohexanol.

- (a) Draw the skeletal formula of menthol and determine the number of chiral centers present in the ring. [2]
-
-
-

- (b) Calculate the maximum number of stereoisomers that could theoretically exist for this molecular formula. [2]
-
-

- (c) Explain why natural (–)-menthol has a fresh peppermint scent, while its synthetic enantiomer (+)-menthol has an unpleasant, medicinal odour. [2]
-
-
-

11

Resolution of Racemic Mixtures Using Diastereomeric Salt Formation — 9701/42/O/N/21/Q7 [6 Marks]

HARD · Syllabus 29.4

A racemic mixture of 2-aminobutane, $(\pm)\text{-CH}_3\text{CH}_2\text{CH}(\text{NH}_2)\text{CH}_3$, cannot be separated by standard fractional distillation.

- (a) Explain why enantiomers cannot be separated by fractional distillation. [2]
-
-

- (b) Describe how adding pure (+)-tartaric acid allows the two enantiomers of 2-aminobutane to be separated. [3]
-
-
-
-

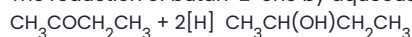
- (c) Explain how the pure amine enantiomers are finally recovered from the separated salts. [1]
-
-

12

Stereochemistry of Carbonyl Reduction by NaBH_4 vs LiAlH_4 — 9701/41/O/N/21/Q7 [6 Marks]

HARD · Syllabus 29.3

The reduction of butan-2-one by aqueous sodium borohydride, NaBH_4 , yields butan-2-ol:



- (a) State the role of NaBH_4 and identify the nucleophile it delivers in this reaction. [2]
-
-

- (b) Explain why the butan-2-ol produced in this reaction shows zero optical rotation. [2]
-
-
-

- (c) Suggest how the reduction of butan-2-one could be modified to yield exclusively (R)-butan-2-ol. [2]
-
-
-

13 Inversion vs Retention in Nucleophilic Substitution Mechanisms — 9701/42/M/J/20/Q7 [6 Marks]

HARD · Syllabus 29.3

A single enantiomer of 2-chlorooctane, $[\alpha]_D^{25} = +36.0^\circ$, reacts with aqueous sodium hydroxide to form octan-2-ol.

- (a) If the reaction proceeds exclusively via an S_N2 pathway, state whether the product will be dextrorotatory, levorotatory, or racemic, explaining the structural inversion. **[3]**

.....

.....

.....

.....

- (b) If the reaction proceeds partially via an S_N1 pathway, predict how the numerical magnitude of $[\alpha]_D^{25}$ will change. **[2]**

.....

.....

- (c) State two solvent conditions that favour S_N2 over S_N1 mechanisms. **[1]**

.....

.....

14 Geometric (cis-trans) vs Optical Isomerism in Cyclic Compounds — 9701/41/M/J/20/Q7 [6 Marks]

HARD · Syllabus 29.1

1,2-Dichlorocyclopropane contains a three-membered ring.

- (a) Draw 3D diagrams of the cis-isomer and trans-isomer of 1,2-dichlorocyclopropane. **[2]**

.....

.....

.....

- (b) State which of the two isomers (cis or trans) exhibits optical isomerism, explaining your choice. **[2]**

.....

.....

.....

- (c) Explain why the cis-isomer is optically inactive despite containing two chiral carbons. **[2]**

.....

.....

15 Stereochemistry of Addition to Unsymmetrical Alkenes: Addition of HBr — 9701/42/O/N/19/Q7 [6 Marks]

HARD · Syllabus 29.3

The electrophilic addition of hydrogen bromide to but-1-ene gives two isomeric bromoalkanes, A and B.

$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 + \text{HBr}$ Product A (major) + Product B (minor)

- (a) Identify Product A and Product B, and explain why Product A is the major product using carbocation stability. **[3]**

.....

.....

.....

.....

- (b) Identify which of the two products (A or B) contains an asymmetric carbon atom, drawing its structure. **[2]**

.....

.....

- (c) Explain why Product A obtained from this reaction does not show any optical rotation. [1]

.....

.....

16 Stereospecific Enzyme Reactions vs Non-Stereospecific Laboratory Reagents — 9701/41/O/N/19/Q7 [6 Marks] **MEDIUM · Syllabus 29.4**

Pyruvate, $\text{CH}_3\text{COCOO}^-$, is converted into lactate, $\text{CH}_3\text{CH}(\text{OH})\text{COO}^-$, both in the human body and in the chemical laboratory.

- (a) In human muscles, lactate dehydrogenase converts pyruvate exclusively into L-lactate. Explain why only one enantiomer is produced. [3]

.....

.....

.....

- (b) In the laboratory, reducing pyruvate with NaBH_4 produces a racemic mixture. Explain this difference in terms of reagent chirality. [3]

.....

.....

.....

SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Definition of Chiral Center & Asymmetric Carbon — 9701/42/M/J/23/Q8 [4 Marks] **MEDIUM · Syllabus 29.1**

Organic molecules often display stereoisomerism.

- (a) Define the term chiral center. [2]

.....

.....

- (b) State which of the following compounds contains a chiral center: pentan-1-ol, pentan-2-ol, pentan-3-ol. [2]

.....

.....

18 Definition of Enantiomers & Mirror Images — 9701/41/M/J/23/Q8 [4 Marks] **MEDIUM · Syllabus 29.1**

Enantiomers are a type of stereoisomer.

- (a) Define the term enantiomers. [2]

.....

.....

- (b) State two physical properties that are identical for a pair of enantiomers. [2]

.....

.....

19 Definition of Racemic Mixture — 9701/42/O/N/23/Q8 [4 Marks]

MEDIUM · Syllabus 29.2

Racemic mixtures are frequently encountered in synthetic chemistry.

- (a) Define the term racemic mixture (racemate). [2]

.....

.....

- (b) Explain why a racemic mixture shows no optical rotation. [2]

.....

.....

20 Stereochemistry of S_N1 Mechanism — 9701/41/O/N/23/Q8 [4 Marks]

MEDIUM · Syllabus 29.3

Tertiary halogenoalkanes hydrolyse primarily by the S_N1 mechanism.

- (a) State the geometry and bond angle around the positively charged carbon in a carbocation intermediate. [2]

.....

.....

- (b) Explain why an S_N1 reaction of a chiral halogenoalkane leads to loss of optical activity. [2]

.....

.....

21 Walden Inversion in S_N2 Mechanism — 9701/42/M/J/22/Q8 [4 Marks]

MEDIUM · Syllabus 29.3

Primary and secondary halogenoalkanes undergo nucleophilic substitution via S_N2.

- (a) Explain what is meant by inversion of configuration. [2]

.....

.....

- (b) Draw the transition state for the reaction of (S)-2-bromobutane with OH⁻, showing partial bonds and negative charge. [2]

.....

.....

.....

22 Number of Stereoisomers of 2,3-Dichlorobutane — 9701/41/M/J/22/Q8 [4 Marks]

MEDIUM · Syllabus 29.1

2,3-Dichlorobutane, CH₃CH(Cl)CH(Cl)CH₃, has two chiral centers.

- (a) Draw the two chiral carbon atoms and explain why it exists as only three stereoisomers. [2]

.....

.....

- (b) Identify which stereoisomer is the meso-isomer and explain why it is optically inactive. [2]

.....

.....

23 Thalidomide Enantiomers & Teratogenicity — 9701/42/O/N/22/Q8 [4 Marks]

MEDIUM · Syllabus 29.4

Thalidomide is a chiral drug administered in the late 1950s.

- (a) Identify the tragic clinical consequence of administering the racemic mixture of thalidomide to pregnant women. [2]

.....

.....

- (b) Explain why administering pure (R)-thalidomide failed to prevent birth defects. [2]

.....

.....

24 Identification of Chiral Centers in 2-Chlorobutane & Butan-2-ol — 9701/41/O/N/22/Q8 [4 Marks]

MEDIUM · Syllabus 29.1

Consider the compounds: 1-chlorobutane, 2-chlorobutane, 2-methylbutan-2-ol.

- (a) Identify which of these compounds is optically active, stating the chiral carbon. [2]

.....

.....

- (b) Draw the two enantiomers of this compound using conventional 3D conventions. [2]

.....

.....

.....

25 Nucleophilic Addition of HCN to Propanal — 9701/42/M/J/21/Q8 [4 Marks]

MEDIUM · Syllabus 29.3

Propanal reacts with HCN to form 2-hydroxybutanenitrile:



- (a) Explain why 2-hydroxybutanenitrile exhibits optical isomerism. [2]

.....

.....

- (b) State and explain whether the product mixture will rotate the plane of plane-polarised light. [2]

.....

.....

26 Chiral Stationary Phase Chromatography — 9701/41/M/J/21/Q8 [4 Marks]

MEDIUM · Syllabus 29.4

Chiral chromatography is used to separate enantiomers in drug manufacture.

- (a) Explain how a chiral stationary phase in HPLC separates a pair of enantiomers. [2]

.....

.....

- (b) State what will be observed in the chromatogram of a racemic mixture. [2]

.....

.....

27 Carbocation Planarity & Loss of Chirality in Dehydration — 9701/42/O/N/21/Q8 [4 Marks]

MEDIUM · Syllabus 29.3

When optically active butan-2-ol is heated with concentrated sulfuric acid, but-2-ene is formed.

- (a) Write the balanced chemical equation for the dehydration of butan-2-ol. [2]

.....

.....

- (b) Explain why the product but-2-ene is optically inactive. [2]

.....

.....

28 Comparison of Stereoisomers: Enantiomers vs Diastereomers — 9701/41/O/N/21/Q8 [4 Marks]

MEDIUM · Syllabus 29.1

Stereoisomerism is divided into enantiomerism and diastereomerism.

- (a) Explain the difference between enantiomers and diastereomers. [2]

.....

.....

- (b) Explain why diastereomers can be separated by standard fractional distillation while enantiomers cannot. [2]

.....

.....

29 Chiral Pool Synthesis Principle — 9701/42/M/J/20/Q8 [4 Marks]

MEDIUM · Syllabus 29.4

Chiral pool synthesis utilizes naturally available enantiomerically pure substances.

- (a) Name two classes of abundant natural products used as starting materials in the chiral pool. [2]

.....

.....

- (b) State one major advantage of chiral pool synthesis over synthesis from petrochemicals. [2]

.....

.....

30 Identification of Asymmetric Carbon in Alanine — 9701/41/M/J/20/Q8 [4 Marks]

MEDIUM · Syllabus 29.1

Alanine, $\text{CH}_3\text{CH}(\text{NH}_2)\text{COOH}$, is an essential building block of proteins.

- (a) Draw the displayed formula of alanine and circle the asymmetric carbon atom. [2]

.....

.....

- (b) List the four different groups attached to the asymmetric carbon in alanine. [2]

.....

.....

31 Optical Rotation of Glucose & Mutarotation — 9701/42/O/N/19/Q8 [4 Marks]

MEDIUM · Syllabus 29.2

When pure α -D-glucose is dissolved in water, its optical rotation changes from $+112^\circ$ to an equilibrium value of $+52.7^\circ$.

- (a) Name this phenomenon where optical rotation changes over time to an equilibrium value. [2]

.....

.....

- (b) Explain why this change occurs in terms of ring opening and closure. [2]

.....

.....

32 Differentiating Enantiomers by Biological Receptors — 9701/41/O/N/19/Q8 [4 Marks]

MEDIUM · Syllabus 29.4

Biological systems are chiral environments that interact selectively with enantiomers.

- (a) Explain why enantiomers taste or smell different (e.g. (+)-carvone smells like caraway, while (–)-carvone smells like spearmint). [2]

.....

.....

- (b) Explain the three-point contact model of drug–receptor interaction. [2]

.....

.....

SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Definition of Plane-Polarised Light — 9701/42/M/J/23/Q9 [2 Marks]

EASY · Syllabus 29.2

Optical activity requires polarised light.

- (a) Define what is meant by plane-polarised light. [2]

.....

.....

34 Condition for Optical Activity in Carbon Compounds — 9701/41/M/J/23/Q9 [2 Marks]

EASY · Syllabus 29.1

Organic molecules may show optical activity.

- (a) State the essential structural feature required for an organic molecule to exhibit optical activity. [2]

.....

.....

35 Definition of Dextrorotatory and Levorotatory — 9701/42/O/N/23/Q9 [2 Marks]

EASY · Syllabus 29.2

Enantiomers rotate light in opposite directions.

- (a) Distinguish between dextrorotatory (+) and levorotatory (–) substances. [2]

.....

.....

36 Formula for Maximum Number of Stereoisomers — 9701/41/O/N/23/Q9 [2 Marks]

EASY · Syllabus 29.1

Molecules can contain multiple chiral centers.

- (a) State the mathematical formula used to calculate the maximum number of stereoisomers for a molecule with n chiral centers. [2]

.....

.....

37 Why Glycine is Not Chiral — 9701/42/M/J/22/Q9 [2 Marks]

EASY · Syllabus 29.1

Glycine is the simplest amino acid.

- (a) Explain why glycine does not exhibit optical isomerism. [2]

.....

.....

38 Definition of a Meso Compound — 9701/41/M/J/22/Q9 [2 Marks]

EASY · Syllabus 29.1

Some molecules with chiral centers are optically inactive.

- (a) Define what is meant by a meso compound. [2]

.....

.....

39 Why Carbocations Lead to Racemisation — 9701/42/O/N/22/Q9 [2 Marks]

EASY · Syllabus 29.3

Carbocations are common intermediates in organic reactions.

- (a) State the shape of a carbocation and explain why it leads to a 50:50 mixture of enantiomers. [2]

.....

.....

40 Meaning of Optically Inactive — 9701/41/O/N/22/Q9 [2 Marks]

EASY · Syllabus 29.2

Polarimeters detect optical activity.

- (a) State what is meant when a liquid sample is described as optically inactive. [2]

.....

.....

SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 [CORE REPEAT 1] Complete Enantiomerism Analysis of Lactic Acid — 9701/42/M/J/23/Q7 [6 Marks]

HARD · Syllabus 29.1

2-Hydroxypropanoic acid (lactic acid) is an optically active carboxylic acid shown in Fig. 41.1.

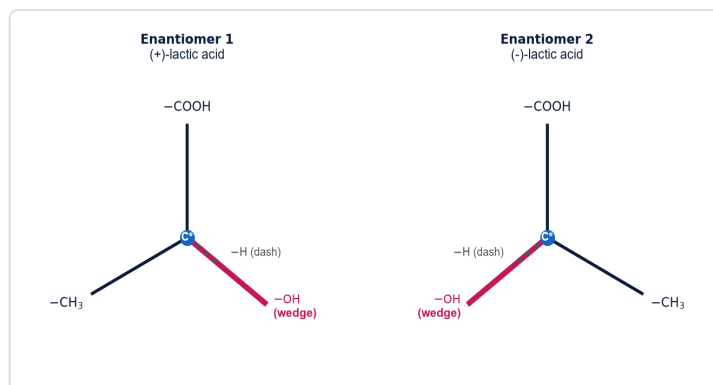


Fig. 41.1: 3D enantiomers of lactic acid showing reflection across mirror plane.

- (a) Explain what is meant by an asymmetric carbon atom, and identify the asymmetric carbon in lactic acid. [2]
-
-
- (b) Draw the 3D structures of both enantiomers of lactic acid reflecting across a mirror plane. [2]
-
-
-
-
- (c) State how the two enantiomers behave when placed in a polarimeter. [2]
-
-

42 [CORE REPEAT 2] Nucleophilic Addition of HCN to Carbonyls & Racemate Mechanism — 9701/41/M/J/23/Q7 [6 Marks] Syllabus 29.3

The reaction of ethanal with HCN/NaCN yields 2-hydroxypropanenitrile.

- (a) Draw the mechanism of nucleophilic addition of cyanide to ethanal, showing all curly arrows and lone pairs. [3]
-
-
-
-
- (b) Explain why the resulting product shows no optical rotation. [2]
-
-
-
-
- (c) State the reagents and conditions to hydrolyse 2-hydroxypropanenitrile into lactic acid. [1]
-
-

43 [CORE REPEAT 3] Stereochemical Pathways: SN1 vs SN2 Hydrolysis — 9701/42/O/N/23/Q7 [6 Marks]

HARD · Syllabus 29.3

Optically active halogenoalkanes exhibit different stereochemical behaviors upon alkaline hydrolysis.

- (a) Explain why S_N2 hydrolysis of a chiral primary halogenoalkane proceeds with complete inversion of configuration. [3]

.....

.....

.....

- (b) Explain why S_N1 hydrolysis of a chiral tertiary halogenoalkane results in racemisation. [3]

.....

.....

.....

44 [CORE REPEAT 4] Chirality in Pharmaceuticals: Single Enantiomers vs Racemates — 9701/41/O/N/23/Q7 [6 Marks] Syllabus 29.4

The anti-inflammatory drug ibuprofen contains a chiral carbon atom.

- (a) Explain why biological enzymes and receptors can distinguish between the two enantiomers of ibuprofen. [2]

.....

.....

- (b) State two disadvantages of selling a chiral drug as a 50:50 racemic mixture. [2]

.....

.....

- (c) Describe how pure enantiomers can be produced using chiral auxiliary synthesis. [2]

.....

.....

45 [CORE REPEAT 5] Identifying Asymmetric Carbons in Aliphatic Chains — 9701/42/M/J/22/Q7 [4 Marks] MEDIUM · Syllabus 29.1

Consider the compound: 3-methylhexane, $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$.

- (a) Identify the chiral carbon atom and list all four groups attached to it. [2]

.....

- (b) Draw 3D wedge-and-dash structures of both enantiomers. [2]

.....

.....

46 [CORE REPEAT 6] Optical Inactivity of Meso Compounds — 9701/41/M/J/22/Q7 [4 Marks] MEDIUM · Syllabus 29.1

Meso-tartaric acid contains two asymmetric carbons but does not rotate polarised light.

- (a) Draw the structure of meso-tartaric acid clearly indicating the internal plane of symmetry. [2]

.....

.....

.....

- (b) Explain why it is optically inactive despite having chiral carbon centers. [2]

.....

.....

47 [CORE REPEAT 7] Planar Carbonyl Attack in Ketone Reduction — 9701/42/O/N/22/Q7 [4 Marks]

MEDIUM · Syllabus 29.3

When phenyl methyl ketone (acetophenone), $\text{C}_6\text{H}_5\text{COCH}_3$, is reduced with NaBH_4 , 1-phenylethanol is formed.

- (a) State why 1-phenylethanol is a chiral molecule. [2]

.....

.....

- (b) Explain why the product obtained from this laboratory reduction is optically inactive. [2]

.....

.....

48 [CORE REPEAT 8] Polarimetry Measurements & Concentration Relationships — 9701/41/O/N/22/Q7 [4 Marks]

MEDIUM · Syllabus 29.2

A pure sample of (S)-lactic acid has a specific optical rotation $[\alpha]_D = -3.8^\circ \text{ dm}^{-1} \text{ g}^{-1} \text{ cm}^3$.

- (a) State the specific optical rotation of pure (R)-lactic acid. [2]

.....

.....

- (b) Calculate the observed optical rotation α when light passes through a 2.0 dm sample tube containing 0.15 g cm^{-3} of (R)-lactic acid. [2]

.....

.....

49 [CORE REPEAT 9] Meaning of Racemisation — 9701/42/M/J/21/Q7 [2 Marks]

EASY · Syllabus 29.3

Chemical reactions can cause racemisation.

- (a) Define the term racemisation. [2]

.....

.....

50 [CORE REPEAT 10] Chiral Center Condition in Alkanes — 9701/41/M/J/21/Q7 [2 Marks]

EASY · Syllabus 29.1

Alkanes can exhibit optical activity.

- (a) Name the smallest alkane that exhibits optical isomerism, giving its molecular formula. [2]

.....

.....

Mark Scheme — Topic 29 — An Introduction to A2 Organic Chemistry

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	CH ₃ -C*(H)(OH)-COOH drawn with asterisk clearly on carbon-2 [1]; Carbon-2 is bonded to four different groups: -H, -OH, -CH ₃ , and -COOH [1].	2
1(b)	Two 3D tetrahedral structures drawn reflecting across a vertical mirror plane with two bonds in plane, one wedge, and one dash [2].	2
1(c)	They rotate the plane of plane-polarised light by equal angles in opposite directions (one clockwise, the other anticlockwise) [1]; Mixing equimolar amounts forms an optically inactive racemic mixture (racemate) where optical rotations cancel out [1].	2
2(a)	Step 1: :CN ⁻ attacks carbonyl carbon with curly arrow from lone pair to C δ^+ , and curly arrow from C=O bond to oxygen [1]; Step 2: Intermediate CH ₃ CH(O ⁻)CN attacks H-CN (or H ⁺) with curly arrow from O ⁻ to H, regenerating :CN ⁻ [2].	3
2(b)	The carbonyl group C=O in ethanal is planar trigonal (120° bond angles) [1]; The nucleophile :CN ⁻ has an equal probability (50:50 chance) of attacking from above or below the plane, producing an equimolar mixture of both enantiomers (racemate) [1].	2
2(c)	The product 2-hydroxy-2-methylpropanenitrile has two identical methyl groups (-CH ₃) attached to the central carbon, so it lacks a chiral carbon center [1].	1
3(a)	Inversion of configuration (Walden inversion) occurs [1]; The OH ⁻ nucleophile attacks the carbon from the rear (180° opposite to the leaving Br atom) [1]; As the five-coordinate transition state breaks, the remaining three groups flip inside out like an umbrella, producing an optically active product of inverted configuration [1].	3
3(b)	Racemisation occurs, resulting in a loss of optical activity / racemic mixture [1]; The slow step involves loss of Br ⁻ to form a planar carbocation intermediate (sp ² hybridized, 120° bond angles) [1]; The nucleophile OH ⁻ can attack the planar carbocation with equal probability from either the top or bottom face, yielding equal amounts of both enantiomers [1].	3
4(a)	Biological receptors, enzymes, and DNA are chiral macromolecules with 3D active sites [1]; Only one enantiomer has the correct spatial orientation of functional groups to bind complementary to the receptor (lock-and-key fit) [1].	2
4(b)	(R)-enantiomer is an effective sedative / alleviates morning sickness in pregnancy [1]; (S)-enantiomer is teratogenic, causing severe foetal limb malformations (phocomelia) [1].	2
4(c)	1. Reduces required dosage by 50%, reducing stress on kidneys/liver [1]; 2. Eliminates adverse side-effects caused by the inactive or toxic opposite enantiomer [1].	2
5(a)	2 ⁿ = 2 ² = 4 stereoisomers [1].	1
5(b)	Meso-tartaric acid drawn with internal plane of symmetry [1]; It possesses an internal mirror plane that bisects the central C-C bond [1]; The optical rotation caused by the upper half of the molecule is exactly cancelled internally by the equal and opposite rotation of the lower half [1].	3
5(c)	Diastereomers are stereoisomers that are not mirror images of each other [1]; Unlike enantiomers (which have identical physical properties), diastereomers have different melting points, boiling points, densities, and solubilities [1].	2
6(a)	Light waves whose electric field vibrations are restricted to oscillate in a single plane [1]; Produced by passing ordinary monochromatic light through a polarising filter (Polaroid sheet or Nicol prism) [1].	2
6(b)	Light source polariser sample tube containing chiral solution analyser detector/eyepiece [1]; The analyser is rotated until minimum/maximum light transmission is observed, reading the angle of optical rotation α ; [1].	2
6(c)	Specific rotation [α] _D = α / (l × c) = +6.5 / (1.0 dm × 0.10 g cm ⁻³) = +65.0° dm ⁻¹ g ⁻¹ cm ³ [1]; If path length is doubled (to 20.0 cm), observed rotation doubles to +13.0° [1].	2
7(a)	Bromine molecule polarised by π -electron cloud; curly arrow from π -bond to δ^+ Br and arrow breaking Br-Br bond [1]; Cyclic bromonium intermediate [CH ₃ -CH(Br ⁺)-CH ₂] formed [1]; Bromide ion Br ⁻ attacks from the opposite face (anti-addition) [1].	3
7(b)	Carbon-2 is chiral (CH ₃ -C*(H)(Br)-CH ₂ Br) [1]; It is bonded to four different groups: -H, -CH ₃ , -Br, and -CH ₂ Br [1].	2
7(c)	Optically inactive [1]; The cyclic bromonium ion can be opened by Br ⁻ attack at either face with equal probability, forming an equimolar racemic mixture [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(a)	Glycine has two identical hydrogen atoms bonded to the central alpha-carbon [1]; Therefore, it lacks an asymmetric carbon with four different groups and has a plane of symmetry [1].	2
8(b)	3D wedge-and-dash structure drawn with alpha-carbon bonded to -H, -NH ₂ , -COOH, and -CH ₃ [2].	2
8(c)	Enzyme active sites are stereospecific 3D chiral pockets constructed exclusively from L-amino acid residues [1]; D-glucose fits snugly into the active site forming precise hydrogen bonds, whereas L-glucose has the opposite spatial geometry and cannot bind [1].	2
9(a)	A chemical reaction that preferentially or exclusively synthesizes one specific enantiomer over the other from an achiral starting material [2].	2
9(b)	1. Chiral resolution: reacting the racemate with a pure enantiomer of another chiral acid/base to form diastereomeric salts with different solubilities, separated by fractional crystallisation [1]; 2. Chiral chromatography: using a stationary phase coated with chiral molecules (chiral HPLC) [1].	2
9(c)	The transition metal possesses bulky chiral ligands (such as BINAP) that create an asymmetric coordination environment around the metal [1]; This sterically blocks one face of the pro-chiral substrate, forcing the incoming reagent to attack exclusively from the open face [1].	2
10(a)	Skeletal formula of cyclohexane ring with -OH at C1, -CH(CH ₃) ₂ at C2, and -CH ₃ at C5 [1]; There are 3 chiral carbon atoms (C1, C2, and C5) [1].	2
10(b)	Number of stereoisomers = $2^n = 2^3 = 8$ stereoisomers [2] (four pairs of enantiomers).	2
10(c)	Olfactory receptors in the nasal epithelium are chiral proteins [1]; (-)-menthol fits precisely into the cold-menthol receptor (TRPM8), eliciting the characteristic cooling peppermint sensation, whereas (+)-menthol binds poorly or triggers different receptors [1].	2
11(a)	Enantiomers have identical intermolecular forces (hydrogen bonding, dipole-dipole, London forces) and therefore identical boiling points [2].	2
11(b)	The racemic base reacts with (+)-tartaric acid to form two salts: [(+)-amine][(+)tartrate] and [(-)-amine][(-)tartrate] [1]; These two salts are diastereomers, not enantiomers [1]; Diastereomers have different solubilities in water/alcohol and can be separated by fractional crystallisation [1].	3
11(c)	Add strong aqueous base (such as NaOH) to liberate the free organic amine, followed by extraction into an organic solvent [1].	1
12(a)	Reducing agent [1]; Delivers the hydride ion, :H ⁻ (nucleophile) [1].	2
12(b)	The carbonyl group C=O in butan-2-one is planar [1]; Hydride attack occurs with equal probability from either side of the flat carbonyl group, producing an equimolar racemic mixture of (R)- and (S)-butan-2-ol whose rotations cancel [1].	2
12(c)	Use an enzyme catalyst (such as alcohol dehydrogenase) or a chiral borane reducing agent (e.g. DIP-chloride / chiral Ru-BINAP catalyst) [2].	2
13(a)	Product will be optically active and inverted (levorotatory, $[\alpha]_D^{25} \approx -36^\circ$) [1]; S _N 2 involves backside nucleophilic attack [1]; The configuration at the chiral carbon is inverted (Walden inversion), reversing the spatial arrangement of bonds [1].	3
13(b)	The optical rotation magnitude will decrease towards zero [1]; S _N 1 generates a planar carbocation that undergoes partial racemisation, diluting the optical purity [1].	2
13(c)	Polar aprotic solvents (such as acetone, DMF, DMSO) / high concentration of strong nucleophile [1].	1
14(a)	cis: both Cl atoms on the same side of the ring plane; trans: Cl atoms on opposite sides of the ring plane [2].	2
14(b)	The trans-isomer exhibits optical isomerism [1]; It lacks any plane or centre of symmetry, meaning its mirror image is non-superimposable [1].	2
14(c)	The cis-isomer possesses an internal plane of symmetry that bisects C3 and the C1-C2 bond [1]; It is a meso compound and is therefore achiral and optically inactive [1].	2
15(a)	Product A = 2-bromobutane; Product B = 1-bromobutane [1]; Protonation of but-1-ene can form a secondary carbocation (CH ₃ CH ₂ C ⁺ HCH ₃) or a primary carbocation (CH ₃ CH ₂ CH ₂ C ⁺ H ₂) [1]; The secondary carbocation is more stable due to the positive inductive (+I) effect of two electron-releasing alkyl groups, forming faster [1].	3

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
15(b)	Product A (2-bromobutane) contains a chiral carbon at C2: $\text{CH}_3\text{-C}^*\text{H}(\text{Br})\text{-CH}_2\text{CH}_3$ [2].	2
15(c)	The secondary carbocation intermediate is planar at the C^+ center; Br^- attacks equally from either face, producing an equimolar racemic mixture [1].	1
16(a)	Lactate dehydrogenase is a chiral enzyme whose active site has a specific three-dimensional geometry [1]; Pyruvate is bound in a fixed, rigid orientation by specific hydrogen bonds and ionic interactions [1]; The coenzyme NADH is held in position to deliver hydride (H^-) exclusively to one specific face (the re-face) of the carbonyl group [1].	3
16(b)	NaBH_4 is a small, achiral reagent in an achiral solution environment [1]; The planar carbonyl group in pyruvate is completely unconstrained and symmetrical to attack [1]; Hydride can collide with equal probability from the top or bottom face, yielding equal numbers of (R)- and (S)-molecules (racemate) [1].	3
17(a)	An atom (usually carbon) bonded to four different atoms or groups of atoms [2].	2
17(b)	Pentan-2-ol [1]; Carbon-2 is bonded to $-\text{H}$, $-\text{OH}$, $-\text{CH}_3$, and $-\text{CH}_2\text{CH}_2\text{CH}_3$ (four different groups) [1].	2
18(a)	Molecules that are non-superimposable mirror images of each other [2].	2
18(b)	Any two from: melting point, boiling point, density, refractive index, solubility in achiral solvents [2].	2
19(a)	An equimolar (50:50) mixture of two enantiomers [2].	2
19(b)	The clockwise rotation caused by one enantiomer is exactly equal in magnitude and cancelled out by the anticlockwise rotation of the other enantiomer [2].	2
20(a)	Planar trigonal [1]; 120° [1].	2
20(b)	The incoming nucleophile attacks the flat carbocation with equal probability from either face [1]; forming equal amounts of both enantiomers (racemisation) [1].	2
21(a)	The spatial arrangement of bonds around the chiral carbon flips to the opposite configuration (like an umbrella turning inside out in the wind) [2].	2
21(b)	Transition state showing central carbon with partial bonds to incoming $\text{HO}\delta^-$ and outgoing $\delta^-\text{Br}$ at 180° , three groups in equatorial plane, square brackets with negative charge [2].	2
22(a)	Both C2 and C3 are bonded to $-\text{H}$, $-\text{Cl}$, $-\text{CH}_3$, and $-\text{CH}(\text{Cl})\text{CH}_3$ [1]; Because the two halves of the molecule are identical, one stereoisomer possesses an internal plane of symmetry (meso) [1].	2
22(b)	(2R, 3S)-isomer is the meso-isomer [1]; It has an internal plane of symmetry so the optical rotations of the two halves cancel internally [1].	2
23(a)	Severe birth defects / limb deformities (phocomelia) / stunted or missing limbs in newborn babies [2].	2
23(b)	The enantiomers interconvert (racemise) rapidly under physiological conditions in human blood [1]; The safe (R)-enantiomer is converted in vivo into the teratogenic (S)-enantiomer [1].	2
24(a)	2-chlorobutane [1]; Carbon-2 is bonded to $-\text{H}$, $-\text{Cl}$, $-\text{CH}_3$, and $-\text{CH}_2\text{CH}_3$ [1].	2
24(b)	Two tetrahedral structures drawn reflecting across a mirror plane with wedge, dash, and in-plane bonds [2].	2
25(a)	Carbon-2 is chiral / bonded to four different groups: $-\text{H}$, $-\text{OH}$, $-\text{CN}$, and $-\text{CH}_2\text{CH}_3$ [2].	2
25(b)	No optical rotation (optically inactive) [1]; Planar carbonyl group is attacked with equal probability from both sides, forming an equimolar racemic mixture [1].	2
26(a)	The two enantiomers interact with the chiral stationary phase with different binding affinities (forming transient diastereomeric complexes) [1]; The enantiomer that binds more strongly is retarded and elutes later, achieving separation [1].	2
26(b)	Two separate peaks of equal area/height (1:1 ratio) [2].	2
27(a)	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3 \rightarrow \text{CH}_3\text{CH}=\text{CHCH}_3 + \text{H}_2\text{O}$ [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
27(b)	But-2-ene does not contain any chiral/asymmetric carbon atoms [1]; It possesses a plane of symmetry (though it exhibits cis-trans geometric isomerism) [1].	2
28(a)	Enantiomers are non-superimposable mirror images; Diastereomers are stereoisomers that are not mirror images of each other [2].	2
28(b)	Diastereomers have different spatial arrangements resulting in different molecular dipole moments and different boiling points [1]; Enantiomers have identical intermolecular forces and identical boiling points [1].	2
29(a)	1. Amino acids (e.g. L-alanine, L-proline) [1]; 2. Carbohydrates / sugars (e.g. D-glucose, tartaric acid) [1].	2
29(b)	The starting material is already 100% enantiomerically pure, avoiding expensive and wasteful resolution of racemic mixtures [2].	2
30(a)	Displayed formula drawn with circle around C2 [2].	2
30(b)	1. Hydrogen atom (-H); 2. Methyl group (-CH ₃); 3. Amino group (-NH ₂); 4. Carboxylic acid group (-COOH) [2].	2
31(a)	Mutarotation [2].	2
31(b)	The cyclic hemiacetal ring opens reversibly to an open-chain aldehyde [1]; The open-chain form recloses with equal ease to form both α-D-glucose and β-D-glucose until equilibrium is reached [1].	2
32(a)	Taste and olfactory receptors are chiral proteins [1]; Only one enantiomer fits into the specific receptor active site to trigger the corresponding nerve impulse [1].	2
32(b)	To distinguish between two enantiomers, a receptor must make contact with the chiral molecule at a minimum of three specific binding sites [1]; Only one enantiomer can align all three groups simultaneously with the matching receptor binding sites [1].	2
33(a)	Light in which all oscillations / wave vibrations occur in a single plane [2].	2
34(a)	Presence of a chiral center / asymmetric carbon atom bonded to four different groups [1]; and lack of an internal plane or centre of symmetry (chiral molecule) [1].	2
35(a)	Dextrorotatory rotates the plane of plane-polarised light clockwise (to the right) [1]; Levorotatory rotates it anticlockwise (to the left) [1].	2
36(a)	Maximum stereoisomers = 2 ⁿ [2] (where n is the number of chiral centers).	2
37(a)	Its alpha-carbon is bonded to two identical hydrogen atoms [1]; It does not have four different groups attached (it is achiral) [1].	2
38(a)	An optically inactive compound that contains two or more chiral centers [1]; but possesses an internal plane of symmetry [1].	2
39(a)	Planar trigonal [1]; Nucleophilic attack occurs with equal probability from either the top face or bottom face [1].	2
40(a)	It does not rotate the plane of plane-polarised light [1]; (either because it is achiral or because it is a racemic mixture / meso compound) [1].	2
41(a)	A carbon atom bonded to four different groups [1]; In lactic acid, C2 is bonded to -H, -OH, -CH ₃ , and -COOH [1].	2
41(b)	Two tetrahedral structures drawn accurately reflecting across a vertical mirror line [2].	2
41(c)	One enantiomer rotates the plane of plane-polarised light clockwise (dextrorotatory) [1]; The other rotates it by the exact same angle anticlockwise (levorotatory) [1].	2
42(a)	Cyanide lone pair attacks carbonyl C ^{δ+} , C=O double bond breaks to O ⁻ [1]; Intermediate alkoxide picks up H ⁺ from HCN (or H ₂ O) to form 2-hydroxypropanenitrile and reform CN ⁻ [2].	3
42(b)	The carbonyl group is planar [1]; Attack by :CN ⁻ occurs with equal likelihood from above and below the plane, producing an equimolar racemic mixture whose opposite rotations cancel [1].	2
42(c)	Heat under reflux with dilute hydrochloric acid (dilute HCl(aq)) [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
43(a)	S _N 2 is a concerted one-step reaction with backside nucleophilic attack [1]; The incoming OH ⁻ attacks the carbon 180° opposite the departing halide [1]; As the leaving group departs, the three remaining groups invert their spatial orientation (Walden inversion) [1].	3
43(b)	S _N 1 proceeds via a two-step mechanism where the slow step forms a flat, planar carbocation intermediate [1]; The nucleophile can attack with equal probability (50:50) from either the front or back face [1]; This produces equal amounts of both enantiomers, resulting in a racemic mixture [1].	3
44(a)	Enzyme active sites are chiral 3D binding pockets made of L-amino acids [1]; Only the active (S)-enantiomer has the correct spatial arrangement of functional groups to bind complementary to the cyclooxygenase (COX) enzyme [1].	2
44(b)	1. Half of the dose is biologically inactive waste, doubling the required dose and increasing metabolic load on liver/kidneys [1]; 2. The inactive enantiomer may cause adverse side-effects or toxicity [1].	2
44(c)	A pure enantiomer of a chiral auxiliary molecule is temporarily attached covalently to the starting material [1]; This forces subsequent reactions to occur stereospecifically from only one face; the auxiliary is then cleaved and recovered, leaving the pure enantiomer product [1].	2
45(a)	Carbon-3 [1]; Attached groups: -H, -CH ₃ (methyl), -CH ₂ CH ₃ (ethyl), and -CH ₂ CH ₂ CH ₃ (propyl) [1].	2
45(b)	Two non-superimposable mirror-image tetrahedral structures drawn with standard wedge-and-dash conventions [2].	2
46(a)	Structure drawn showing dashed mirror plane bisecting the central C-C bond with identical top and bottom halves [2].	2
46(b)	Internal compensation: the optical rotation caused by one chiral center is exactly equal and opposite to the rotation caused by the other chiral center within the same molecule [2].	2
47(a)	The central carbon is bonded to four different groups: -H, -OH, -CH ₃ , and -C ₆ H ₅ [2].	2
47(b)	The C=O group of acetophenone is planar; hydride attack from above and below occurs at equal rates, giving an equimolar racemic mixture [2].	2
48(a)	$[\alpha]_D^{25} = +3.8^\circ \text{ dm}^{-1} \text{ g}^{-1} \text{ cm}^3$ [2] (equal magnitude, opposite sign).	2
48(b)	$[\alpha]_D^{25} = [\alpha]_D^{25} \times l \times c = +3.8 \times 2.0 \times 0.15$ [1]; $[\alpha]_D^{25} = +1.14^\circ$ (or $+1.1^\circ$) [1].	2
49(a)	The conversion of an optically active single enantiomer into an optically inactive equimolar racemic mixture [2].	2
50(a)	3-methylhexane [1]; C ₇ H ₁₆ [1].	2